Methodology Passport Card

AI-Based Adaptive Learning in Higher Education — Task 2 (revised)

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1. Project information

Field	Value
Title	The Effectiveness of Artificial Intelligence-Based Adaptive Platforms on Student Academic Performance in Higher Education
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GitHub repository	https://github.com/nurba-git/research-methods
Course / deliverable	Research Methods — Task 2 (Methodology Card & Repository), revised

2. Research type and design

Research type: quantitative, confirmatory, experimental. The research question asks whether one learning condition *causes* a larger, measurable change in test scores than another. That calls for numerical outcomes, a pre-specified hypothesis test and a design that supports causal inference, so a qualitative or mixed-methods approach is not chosen for the primary question.

Design: two-arm, parallel-group randomized controlled trial (RCT). The design is fixed; it is not left open between an RCT and a quasi-experiment.

Design element	Decision
Unit of randomization	Individual student
Arms	**Baseline:** traditional learning in the university LMS · **Treatment:** the same course module delivered through an AI-adaptive platform
Allocation	1:1, stratified by pre-test tercile (low / middle / high), computer-generated sequence with fixed seed, concealed from instructors until enrolment closes
Duration	Week 0 pre-test → 8 weekly learning sessions → Week 9 post-test
Warm-up	Sessions 1–2 are onboarding and cold-start calibration of the adaptive model. They are logged but excluded from session-level metrics
Blinding	Students cannot be blinded; post-tests are scored automatically or by a marker blind to arm
Planned sample	N = 140 (70 per arm). 64 per arm detect d = 0.5 at α = 0.05 (two-sided), power = 0.80; +10% for attrition
Analysis population	Complete cases (randomized and post-tested); attrition reported per arm

3. Research question

RQ1 (causal). How does learning with an AI-based adaptive platform affect university students' academic score gain and time on task, compared with traditional learning on equivalent material?

RQ2 from Task 1 (which algorithm parameters predict learning difficulties best) is deferred to a later technical study; the current experiment does not manipulate algorithm parameters.

4. Hypotheses

- **H_0 (null):** The mean score gain (post-test − pre-test) of the adaptive arm equals that of the traditional arm: μ(adaptive) − μ(traditional) = 0.
- **H_1 (alternative):** The mean score gain differs between arms: μ(adaptive) − μ(traditional) ≠ 0 (expected direction: adaptive > traditional).

Decision rule: reject H_0 if the two-sided Welch t-test gives $p < 0.05$ **and** the 95% bootstrap CI of the mean difference excludes 0. Effect size is reported as Hedges' g. An ANCOVA (post-test ~ arm + pre-test) is run as a pre-specified sensitivity analysis.

5. Variable matrix

Category	Variable	Operational definition	Unit / levels
Independent	Learning approach	Arm assigned by randomization	`traditional` (baseline), `adaptive`
Dependent — primary	Score gain	Post-test score − pre-test score on parallel test forms	points (−100 … 100)
Dependent — secondary	Time on task	Mean minutes per weekly session (evaluation sessions 3–8)	minutes
Dependent — secondary	Session quiz score	Weekly quiz score	% correct
Controlled	Baseline knowledge	Same pre-test; stratified randomization; pre-test as ANCOVA covariate	points
Controlled	Content and assessment	Same learning objectives, same item bank, same scoring rubric in both arms	–
Controlled	Schedule and environment	Same course, semester, session length and LMS access conditions	–
Controlled	Analysis environment	Python 3.11.11, pinned packages (`requirements.txt`), Docker image, seed = 42	–
Recorded covariates	Study programme, prior experience with AI platforms	Collected only with consent	categorical

6. Metrics and baseline

Primary metric

Difference in mean score gain, Δ = mean gain(adaptive) − mean gain(traditional), in points, with a 95% bootstrap CI (10 000 resamples) and Hedges' g.

Secondary metrics

- Difference in mean time on task (minutes; negative = adaptive faster).
- Weekly quiz score trajectory per arm (learning curve), sessions 3–8.

Guardrail metrics

Guardrail	Threshold	Why
Platform error rate	≤ 2% of evaluation sessions with `status = error`	A broken platform would bias the comparison
Platform response latency	p95 ≤ 1000 ms in evaluation sessions	Slow feedback harms the learning experience
Attrition	≤ 15% per arm, and similar between arms	Differential drop-out breaks randomization
Data validity	Scores in 0–100, times > 0; invalid rows reported, never silently dropped	Measurement integrity

Baseline (control group)

Traditional learning: the same module, readings, exercises and weekly quizzes delivered as static pages in the university LMS, in a fixed order, with no algorithmic adaptation. The adaptive arm receives the same item bank, but the platform chooses item order and difficulty from each student's predicted mastery.

7. Procedure and reproducibility

1. Obtain ethics approval and informed consent; pseudonymise student IDs.
2. Week 0: administer the pre-test; randomize within pre-test terciles.
3. Weeks 1–8: one learning session per week; the platform logs one JSON line per session.
4. Week 9: administer the post-test.
5. Pre-register the primary outcome, test and decision rule before unblinding.
6. Run `src/process_data.py` and `src/visualize.py` (or the notebook) with the versioned `configs/config.yaml`; outputs go to `data/processed/` and `reports/figures/`.

8. Threats to validity

Threat	Type	Mitigation
Selection bias	Internal	Individual randomization, stratified by pre-test
Novelty / cold-start effect	Internal	Sessions 1–2 treated as warm-up; outcome measured at Week 9
Contamination between arms	Internal	Separate course shells; students asked not to share platform access
Instructor or content differences	Internal	Identical objectives, item bank and rubric in both arms
Differential attrition	Internal	Attrition guardrail; report per arm; sensitivity analysis
Test practice effect	Construct	Parallel pre- and post-test forms
Small or single-site sample	External / statistical	A-priori power analysis; report CIs; replicate in other courses
Synthetic demonstration data	Status	The repository currently runs on simulated data; no empirical claim is made

9. Repository status

The repository contains a tested, containerised pipeline. `src/benchmark.py` runs a smoke test on a 10-row sample. The Task 3 analysis (`src/process_data.py`, `src/visualize.py`, `notebooks/01_data_analysis.ipynb`) currently runs on a **simulated** session log generated by `src/simulate_data.py` with seed 42. It will be re-run unchanged on the real log once collected.